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Chapter 4 The Integers

How do we define negatives?

→ We will do it by thinking of differences between sets.

$x+3=2$, $x^2=4$

Nice for solving equations!

Say B has size 4 & A " " 3.

$(3,4) \approx -1$

Def. 1

Let $\Delta : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ be defined by

$$\Delta(m,n) = \begin{cases} (m-n, 0) & \text{if } m \geq n \\ (0, n-m) & \text{if } n > m \\ (0,0) & \text{if } n = m \end{cases}$$

← negative case

This is well defined by trichotomy.

we have

we owe

$\Delta(4,3) = (1,0)$ → "equals pos. 1"

$\Delta(3,4) = (0,1)$ → "equals minus 1"

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building h^{-1}:

I know h

$\to$ is 1 to 1

$\to$ is onto

If $a \in S$,
Define $h^{-1}(a)$ to be the $k \in \{1, \dots, n\}$ st. $h(k) = a$.

Why is this reasonable? [Why is h^{-1} well defined?]

$\downarrow$

First, since h is onto for every $a \in S$, $\exists\ k \in \{1, \dots, n\}$ such that $h(k) = a$.

Second, since h is injective (1 to 1), if $h(k) = h(l) = a$ then $k = l$.

§3.3 New Perspective on Addition

$\downarrow$

<u>Alternative Def</u>:
Let $m, n \in \mathbb{N}$, define $m + n \overset{\text{def}}{=} |A \cup B|$ where A & B are disjoint sets with $|A| = m$ & $|B| = n$.

Associativity $\to$ $A \cup B = B \cup A$

$\boxed{\text{Thm. 21}}$

Every subset of a finite set is finite.

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Thm. 21

Every subset of a finite set is finite.

Thm. 22

If $A \subseteq C$, then $|A| \leq |C|$.
If $A \subsetneq C$, then $|A| < |C|$.

Sketch:

Consider $C - A = \underbrace{C \setminus A}_{\text{everything not in } C}$

Let $b = |C - A|$

If we let $|A| = a$ & $|C| = c$ then we see
$\underbrace{C = A \cup (A \setminus C)}$,

C = everything in A & not in A

So $|C| = |A \cup (C - A)|$ thus $c = a + b$.

By Ass 2 $\overline{\text{with equality}}$ iff $b = 0$

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Thm. 22

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So $|C| = |A \cup (C - A)|$ thus $c = a + b$.

By Chp. 2, $a \leq c$ with equality iff. $b = 0$.

$\square$

Thm. 23

Let $f: A \to B$ be an injection (1 to 1).

Suppose $|B| = b < \infty$. Then $|A| = a < \infty$ and $a \leq b$.

If f is not surjective, then $a < b$.

→ this basically gives a copy of A onto B.

Pf

Consider $f(A) = \cancel{\{b \in B : f(a) = b}$ for some $a \in A\} \subseteq B$.

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→ this basically gives a copy of A onto B.

<u>Pf</u>

Consider $f(A) = \{b \in B : f(a) = b \text{ for some } a \in A\} \subseteq B$.

Let $g : A \to f(A)$ given by $g(a) = f(a)$.

Note g is injective & surjective (by def. of $f(A)$).

Thus g is bijective.

By thm. 10, $|A| = |f(A)| \overset{\text{Thm. 21}}{=} a < \infty$.

* Applying thm. 22 to $f(A) \subseteq B$, we see $a = |A| \leq |B| = b$.

If f is not surjective, then $f(A) \subsetneq B$ and thm. 22 says $a < b$.

$\square$

<u>Contrapositive</u>

Not ① ⟹ Not ②

$|A| > |B| \Rightarrow f : A \to B$ is not injective

* Pigeonhole principle: A = pidgeon set B = hole set

$\exists\, b \in B$ st. $f(n) = b = f(c)$ & $a \neq c$.

§3.5 New Perspective on Mult.

<u>The Cartesian Product</u>

of two sets A & B is $A \times B \overset{\text{def}}{=} \{(a, b) : a \in A, b \in B\}$

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§3.5 New Perspective on Mult.

The Cartesian Product

of two sets A & B is $A \times B \overset{\text{def}}{=} \{(a,b) : a \in A, b \in B\}$

ex

xy plane is $\mathbb{R} \times \mathbb{R}$.

For $m, n \in \mathbb{N}$ define $m \times n = m \cdot n = mn \overset{\text{def}}{=} |A \times B|$
where $|A| = m$ & $|B| = n$

$2 \times 3 = |A \times B|$ where $A = \{\bigcirc, \square\}$
$B = \{\star, \triangle, \diamond\}$

$$A \times B: \begin{cases} (\bigcirc, \star), (\bigcirc, \triangle), (\bigcirc, \diamond) \\ (\square, \star), (\square, \triangle), (\square, \diamond) \end{cases}$$

§3.7 New Perspective on Exponentiation

DEF:
For sets A & B define B^A to be the set of functions from A to B.

Exponentiation: $|A| = a$ & $|B| = b$, then $|B^A| = b^a$

ex

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§3.7 New Perspective on Exponentiation

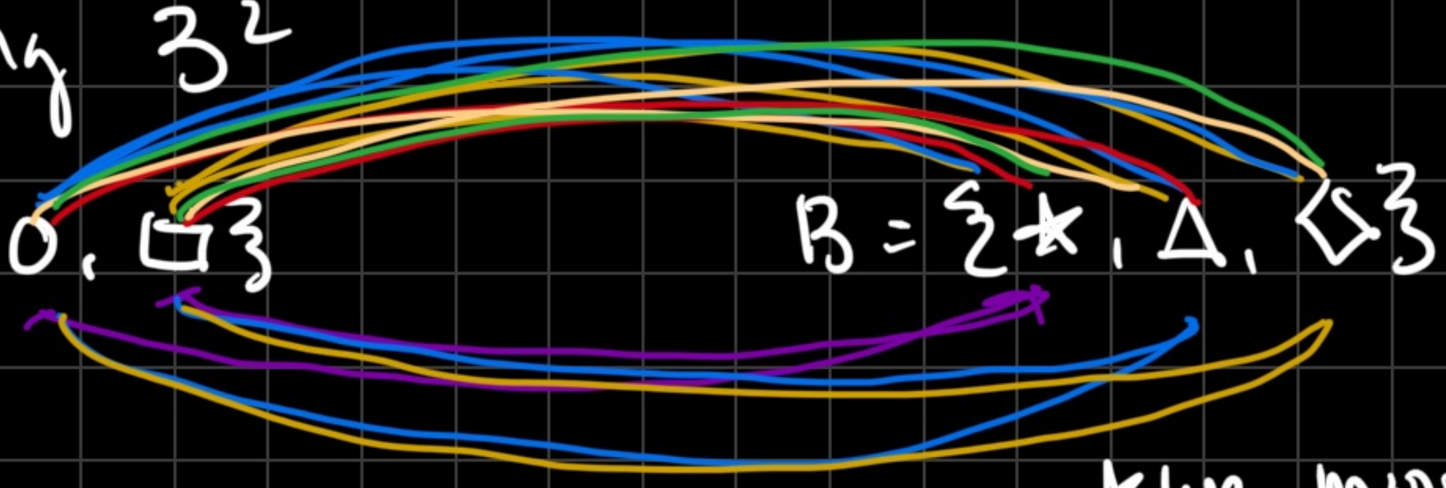
DEF:
For sets A & B define B^A to be the set of functions from A to B.

Exponentiation: $|A| = a$ & $|B| = b$, then $|B^A| = b^a$

ex

Computing 3^2

$A = \{\bigcirc, \square\}$ $B = \{\star, \triangle, \diamond\}$



★ we mapped 9 times

- exponentiality is just counting functions!